

THESHIKA NAVOD | COMPUTER ENGINEERING

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PROFILE SUMMARY

AI Engineer specializing in **Generative AI, Multi-Agent Systems, and Prompt Engineering**. Demonstrated expertise in building **production-grade RAG systems, agentic architectures, and enterprise AI solutions**. Seeking an AI Engineering role to design scalable intelligent systems.

WORK EXPERIENCE

Applantics (PVT) LTD Associate Software Engineer

February 2025 – Current

Developing AI-integrated POS system with predictive analytics, inventory management, and real-time transaction processing. Building scalable microservices, integrating payment gateways, and developing analytics dashboards for business insights. Developing RAG-powered AI chatbots with LangChain and vector databases.

Technologies: Java, JavaScript, Python, React, MySQL, MongoDB, AI Agents, AI Chatbots, LangChain, Docker, AWS.

Information Systems Associates (ISA) Intern Software Engineer

July 2024 – January 2025

Developed microservices for Air Arabia's aviation systems using Spring Boot, React, Kafka, PostgreSQL. Refactored codebase resolving 200+ critical issues, increased test coverage to 80%, implemented real-time boarding pass features, converted REST to gRPC, deployed Docker containers via Jenkins CI/CD.

Technologies: Java (8, 11, 17), Spring Boot, React, Angular, PostgreSQL, Oracle DB, Redis, Kafka, gRPC, Docker, Jenkins, JUnit, Mockito, SonarQube, Git, SOAP/REST APIs, Agile environment with Jira/Confluence

PROJECT EXPERIENCE

AI Code Generator and Reviewer System - Research Project (code - upon request)

Built VS Code extension automating code generation and bug detection for legacy codebases, achieving 42% score on SWE-bench. Developed multi-agent reviewer system using fine-tuned LLMs detecting security vulnerabilities, code smells, and performance issues with human-in-the-loop validation, reducing manual review time by 60%.

Technologies: Python, TypeScript, LangGraph, LLM APIs, Multi-Agent system, Agentic RAG, AST, VS Code Extension API, Fine-tuning, Prompt engineering, Transformer Models, Pinecone.

Dataflow RAG-Based Customer Service Agent

Built an enterprise grade RAG system for automated customer support using local AI models. Achieved semantic search accuracy through FAISS vector store (knowledge base) with optimizing chunks and embeddings.

Technologies: Python, LangChain, FAISS, HuggingFace Embeddings, RAG Architecture, Vector Databases

Business Intelligence Multi-Agent Analysis System

Built an automated business analysis system using CrewAI framework with 3 agent workflow (Data Collector, Analyst, Report Writer). Automated sales trend identification and generated actionable business recommendations.

Technologies: Python, CrewAI, OpenAI, Multi-Agent Systems, Data Analysis, Report Generation

Multi Agent Customer Support System (Support Flow)

Multi-Agent system for intelligent ticket triage, solution research, response generation. Implemented sequential, parallel workflows handling complex multi-issue tickets. Deployed locally for data privacy, zero external API dependencies.

Technologies: Python, Microsoft AutoGen, Ollama, Multi-Agent Architecture, LLM Orchestration

Sinhala Sign Language Learning Management System

Developed AI-powered LMS for learn Sinhala Sign Language. Achieved 87% gesture recognition accuracy in real-time hand gestures using computer vision, MediaPipe pose estimation, and deep learning models.

Technologies: Python, Computer Vision, OpenCV, MediaPipe, Deep Learning, TensorFlow, FastAPI, NLP

Jenkins pipeline for a Spring Boot application

Pipeline: Checkout, Test & Build, SonarQube Analysis, Quality Gate, Package, Docker Build, Deploy, and monitoring.

Tech Stack: Java 21, Spring Boot, PostgreSQL, Jenkins, Docker, SonarQube, Prometheus, Grafana, Gradle

HPC Optimization for LLM Parallel Prompt Processing

Optimized LLM prompt processing using HPC techniques achieving 3.27x speedup. Implemented and benchmarked OpenMP, MPI, and hybrid parallelization strategies across Pure Python, Pure C, and C + Python processing prompts.
Technologies: C, Python, OpenMP, MPI, HPC Techniques, Performance Optimization, API Integration

High-Security Voting System

Developed highly secure voting system with cryptographic authentication, end-to-end vote encryption, and tamper-proof architecture ensuring anonymous, verifiable elections

Technologies: Java, Spring Security, JWT, Role-based Auth, Multi Factor Auth, Email Verification, Cryptography

Enterprise Microservices Platform

Production-grade microservices with API Gateway, service discovery, event-driven architecture, OAuth2 security, circuit breaker, distributed tracing, and monitoring. Containerized deployment.

Technologies: Java Spring Boot, Spring Cloud Gateway, Netflix Eureka, Kafka, Keycloak, Docker, Grafana

EDUCATION

B.Sc. (Hons) Computer Engineering.
Faculty of Engineering, University of Ruhuna
GPA: 3.30

CERTIFICATES

[Mastering AI Agents Bootcamp: Build Smart Chatbots & Tools](#) - School of AI, Udemy
[Mastering Context Design for Intelligent AI Agents](#) - Vivian Aranha, School of AI, Udemy
[From Prompt Engineering to Agent Engineering](#) - Anton Voroniuk, Anton Voroniuk, Eugene Vyborov, Dmytro Vasyliiev
[Certified Generative AI Architect with Knowledge Graphs](#) - Vivian Aranha, School of AI, Udemy
[Building Effective Agentic Systems with Generative AI](#) - Dr. Amar Massoud, Udemy
[Model Context Protocol \(MCP\) Course](#) - Hugging Face
[Java: Complete Java Core for Beginners with Exercises - 2025](#) - Andrii Partola, ITER Academy
[Spring Framework for Java Developers: Practical Guide](#) - Andrii Piatakha, Udemy
[CI/CD with Jenkins and Docker](#) - School of Devops, Gourav J. Shah Udemy

TECHNICAL SKILLS

- **AI/ML:** LangChain, AutoGen, CrewAI, RAG Architecture, Multi-Agent Systems, LLM Integration, Vector Databases, HuggingFace, Prompt Engineering, Computer Vision, Deep Learning, Fine-tuning, MLOps
- **Programming Languages:** Python, Java, TypeScript, JavaScript
- **Backend:** Spring Boot, Spring Cloud, Node.js, NestJS, FastAPI, Apache Kafka, RabbitMQ, Junit, Unit Testing
- **Frontend:** React, Next.js, Angular, HTML5, CSS3, Tailwind CSS
- **Databases:** PostgreSQL, MySQL, MongoDB, Redis, Neo4j, Pinecone
- **DevOps & Cloud:** Docker, Kubernetes, Jenkins, AWS, CI/CD
- **Architecture:** Microservices System Design, RESTful APIs, gRPC, Event-Driven Systems, Server-Sent Events

SOFT SKILLS

Communication & Presentation | Stress Management | Time Management | Team Collaboration | Leadership

INVOLVEMENT

- 3rd Year Field Representative – Computer Engineering
- IEEE INSL Competition – participation
- Leadership Development program - Maliyadeva College
- Haxtreme 2.0 - 4th place
- Former Senior Prefect – Mayurapada Central College
- Football Team Champion – Mayurapada Central

REFERENCES

Dr.Kushan Sudheera - Senior Lecturer, Faculty of Engineering, University of Ruhuna.. +94719693164, kushan@eie.ruh.ac.lk
Dr. Rajitha Udawalpola - Senior Lecturer, Faculty of Engineering, University of Ruhuna. + 94718578608, rajitha@eie.ruh.ac.lk